

ARCHIT AGRAWAL

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EXPERIENCE

AI Software Engineer — EdPlus, Arizona State University Tempe, AZ | Jun 2026–Present

- Replaced bespoke per-survey pipelines with a reusable platform of 10 agents and 65 registered tools, orchestrated from a declarative graph with TypeScript, Bedrock, AgentCore, Step Functions, and CDK.
- Prevented roughly 12% silent row loss through schema-constrained decoding, response correlation tokens, and citation verification; achieved 100% join integrity and zero label churn across repeat runs.
- Made newly uploaded surveys immediately queryable without dataset-specific configuration by building a natural-language semantic layer over DuckDB.
- Redirected the team roadmap through a controlled evaluation showing model choice changed labeling F1 by 0.230, compared with 0.007 for pipeline architecture.
- Eliminated a Distributed Map race through partitioned writes and fan-in, verified across 258 runs; held query latency flat at 200× row volume with DuckDB over Lance on S3.

Founding AI/ML Engineer — MyStage Music Inc. Remote | Jul 2025–Present

- Reduced service dependencies from 14 to 5 by replacing a 14-Cloud-Function pipeline with one checkpointed LangGraph worker on Cloud Run.
- Indexed 70,000+ records daily across 650+ venues through a Playwright service; improved search accuracy 25% with Gemini entity resolution and deduplication.
- Served search at sub-50 ms latency through FastAPI and Algolia with real-time Firestore synchronization.
- Made concurrent retries safe with atomic claims and deterministic task IDs; added human-in-the-loop pause/resume, Logfire tracing, and 200+ pytest tests.

Software Engineer — EdPlus, Arizona State University Tempe, AZ | Sep 2023–May 2025

- Supported 1,000+ faculty authoring courses for 60,000+ students by leading development of a multi-tenant RAG assistant.
- Reduced transcript analysis from four hours to 15 minutes with a Neo4j knowledge graph and schema-constrained, LLM-generated Cypher.
- Reduced college onboarding from a deployment to a configuration change while enforcing tenant isolation in retrieval; evaluated groundedness, relevance, and coherence.

Data Research Aide — Knowledge Exchange for Resilience, ASU Tempe, AZ | Jun 2024–Aug 2024

- Reduced p95 retrieval latency 60% at 95% recall with hybrid FAISS and Neo4j retrieval; reduced API latency from 198 ms to 20 ms through Redis and SQL optimization.
- Delivered 20+ .NET 8 REST APIs for an NSF-funded platform and an idempotent ETL pipeline ingesting 10,000+ faculty profiles daily.

Software Engineer — Zeus Learning Mumbai, India | Jan 2022–Jul 2023

- Reduced resource usage 35% and infrastructure cost 20% by migrating a .NET monolith to Kubernetes microservices through a strangler pattern.
- Reduced deployment time 70% and incidents 40% with SonarQube merge gates and containerized delivery; shipped demand prediction across 300+ enterprise sites.

Product Intern — EAT.FIT Bengaluru, India | Sep 2021–Dec 2021

- Reduced driver-location serving cost 45% through distance-aware polling and client interpolation; built a backpressure-aware WebSocket notification service.

EDUCATION

M.S., Computer Science, Arizona State University — GPA: 4.0 Aug 2023–May 2025

B.Tech., Information and Communication Technology, DA-IICT Aug 2018–May 2022

SELECTED PROJECTS

MCP GitHub PR Review Agent: Built a webhook-triggered MCP service combining repository state, ticket metadata, and CI context for automated review decisions.

SRP Electric MCP Server: Exposed an undocumented utility portal as validated TypeScript tools after reverse-engineering its authentication flow.

Video Generation Physics Evaluation: Built an evaluation harness separating physical plausibility from semantic fidelity; found counterfactual prompts degraded faster at matched length.

SKILLS

Languages: Python, TypeScript, C#, SQL, JavaScript

AI & Data: LangGraph, MCP, Bedrock/AgentCore, Vertex AI, OpenAI, RAG, DuckDB, PostgreSQL/pgvector, Neo4j, FAISS, Lance, Algolia, Redis, Firestore

Backend & Cloud: FastAPI, NestJS, .NET 8, React, Nuxt 4, AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions

Quality: pytest, Vitest, Locust, Logfire, distributed tracing